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ORIGINAL PAPER

Cancer among Asian Indians/Pakistanis living in the United States: low incidence and generally above average survival

William B. Goggins · Grace Wong

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Abstract
Background South Asian immigrants living in the United Kingdom and Canada have been found to have lower rates of cancers of all types compared with the native born population and most other immigrant groups. Cancer among Asian Indian/Pakistani people in the United States has been studied very little.
Methods Incidence rates for all cancers combined and site-specific rates for major cancers were estimated for Asian Indians/Pakistani population using incidence data from the U.S. National Cancer Institutes SEER database and population data from the U.S. Census Bureau. Site-specific survival was compared for major cancer sites between Asian Indians/Pakistanis and Caucasians using Cox proportional hazards models.
Results Cancer rates for Asian Indian/Pakistani males and females were considerably lower than for White Americans with standardized incidence ratios (SIRs) of 0.46 (95% CI = 0.44, 0.48), and 0.55 (95% CI = 0.53, 0.58) respectively. Site-specific rates were lower for both genders for most sites with particularly low rates observed for lung, colorectal, female breast, and prostate cancer. Among common cancers sites, survival was generally better among Asian Indians/Pakistanis than Caucasians with the notable exception of breast cancer for which Caucasians had slightly better survival.
Conclusions The finding that Asian Indians/Pakistanis in the United States have relatively low incidence rates for most major cancers is consistent with studies from other countries. Whether the low incidence of cancer and above average cancer survival for this group is related to their well-above average socioeconomic status or cultural and behavioral factors is a topic for further research.

Keywords SEER · Ethnic · Race · South Asian · Inequality

W. B. Goggins (✉) · G. Wong
School of Public Health, Chinese University of Hong Kong, Room 501, Shatin, Hong Kong
e-mail: wgoggins@cuhk.edu.hk

W. B. Goggins
Nethersole School of Nursing, Chinese University of Hong Kong, Shatin, Hong Kong

Introduction

Considerable ethnic variation in the incidence and survival rates for many types of cancers have been reported from the United States and other countries. One group whose cancer experience has been little studied in the United States are Asian Indians, a mostly affluent, and well-educated group whose numbers have been rapidly increasing in recent years.

Previous studies have shown that cancer rates for South Asians are generally lower than those of the non-South Asian population of that country but higher than those of the Indian subcontinent. A study using data from the California cancer registry found that South Asians, of whom about 90% were Asian Indians, had lower cancer incidence than non-Hispanic whites for most sites, but higher incidence than Whites for liver cancer for both genders, and stomach, esophageal, and cervical cancer for females [1]. A registry-based study from the United Kingdom [2] found that South Asians had substantially lower incidence rates for most major cancers than non-South Asians. A Canadian study [3] found that cancer mortality rates among South Asians were lower than those of both White and Chinese Canadians. Data from Singapore indicate that Indians there recently have lower cancer incidence than either Chinese or Malays [4], whereas incidence rates for Indians in Malaysia are lower than those for Chinese but higher than those for Malays [5].

A study of female breast cancer survival using U.S. cancer registry data found that Asian Indian/Pakistani women had non-significantly worse overall survival than non-Hispanic Whites after adjustment for demographic variables only, but that they had non-significantly better survival after additional adjustment for disease characteristics [6]. Two studies from the United Kingdom [7, 8] have found that South Asian women, and another found that they had about the same survival rate [9]. A study of colorectal cancer survival using U.S. cancer registry data found that Asian Indians/Pakistanis had significantly better survival compared with Whites [10].

In this article, we present findings on cancer incidence and survival on this group using data from the U.S. National Cancer Institute's (NCI) Surveillance Epidemiology and End Results (SEER) program. Because Asian Indians and Pakistanis are included as one category in the SEER data these groups are analyzed together. However, as about 94% of the Asian Indian/Pakistani population of the relevant SEER areas is Asian Indian, the inferences drawn from this study mostly apply to this group.

Methods

Incidence

Incidence rates for cancers for Asian Indians/Pakistanis were estimated using data from the seven SEER registries with the largest populations for these groups: the state of Connecticut, and the cities of San Francisco, San Jose, Los Angeles, Detroit, Seattle, and Atlanta. The gender- and age-specific counts of malignant neoplasms during 1988 to 2004 were obtained using SEER's SEERStat 6.3.5 program. The number of person-years was estimated using data from the U.S. Census bureau. First, the Asian Indian and Pakistani populations for the relevant SEER areas were obtained for 1990–2000, and then linear interpolation was used to estimate the number of person-years. This method assumes that the population change for the period from 1988 to 2004 was constant, and the yearly rate of population change was equal to the difference between the 2000 and 1990 populations divided by 10. Gender- and age-specific rates for cancer overall and for common sites were then calculated as cases divided by person-years. Age-standardized rates (ASR) were estimated using the world standard population, and standardized incidence ratios (SIRs) comparing the rates for South Asians against rates for Whites were then calculated in the standard fashion using the standard method [11]. Confidence intervals (CIs) for the SIRs were calculated using the "errors-in-variables" approach [12] because the reference rates and person-years used were not exact. The reference rates for Whites were calculated using data from the same six registries only. ASRs for the Indian registries in Bombay and Madras were obtained from the IARC's Cancer Mondial Statistical Information System [13].

Survival

Data from all the 17 SEER registries were used for the survival comparisons. Analysis was restricted to 1988–2004 because Asian Indians/Pakistanis were classified as "Other Asians" before 1988. Site-specific survival was analyzed for cancers of the colon, rectum, stomach, esophagus, pancreas, liver, lung and bronchus, prostate, female breast, ovaries, brain, and non-Hodgkin's lymphoma (NHL). Cox proportional hazards models were used to estimate relative risks (RRs) and their 95% CIs for all-cause and cause-specific mortality for each of the above sites after adjusting for relevant confounders. For the cause-specific mortality analyses only deaths from cancer of the site being analyzed were the considered events, and subjects were considered censored at time of death if they had died of other causes. Relative risks were estimated for South Asians and Chinese with non-Hispanic Whites serving as the reference group. Results for Chinese, another recent immigrant group with an above-average SES profile, were included for comparison purposes. The proportional hazards assumption was verified by visual examination of Kaplan–Meier survival plots. The correct functional form for continuous covariates, age and year of diagnosis, was determined by examining scatterplots of Martingale residuals from Cox models fit omitting the relevant covariate vs. that covariate [14]. Two models were fit for each type of mortality, one adjusting only for demographic factors including age (using linear and quadratic terms), gender, year of diagnosis, and registry (through stratification). For solid tumors, the other models additionally adjusted for disease characteristics include, SEER historic stage (local, regional, distant, or unstaged), tumor size (categorized into quartiles), grade (I, II, III, IV, or no grade), and histology. For histology, separate categories were created for more common histologies (more than 5% of cases) with other types being grouped together as "others". Models for NHL survival were adjusted for SEER extent of disease, grade, and histology. All disease variables were controlled for using stratification in order to avoid any problems due to non-proportional hazards. Analyses were restricted to malignant cases. All tests were two-tailed and p-values < 0.05 were considered statistically significant.

Results

Incidence

Age-standardized incidence rates (ASR) for Asian Indians/Pakistanis, SIRs comparing their incidence rates to those of Whites living in the same geographical areas, and rates from the Madras and Bombay Indian registries are shown in Table 1. Overall cancer incidence was around 50% lower in Asian Indians/Pakistanis than in the White population, but higher than those observed for the Indian registries. Particularly low rates (relative to the U.S. White population) were observed for smoking-related sites, including lung and bronchus (SIRs = 0.29 for males and =0.20 for females), larynx, and urinary bladder. Substantial but non-significant excess risk (compared with U.S. whites) was observed only for esophageal cancer in females and gum and other mouth cancer for males. Compared to the Indian registry rates large increased risks were noted for colorectal and prostate cancer, whereas large decreases are found for stomach and oral cavity cancers.

Survival

Relative risks, 95% CIs and p-values for the four types of models by site are given in Table 2. Results by site are discussed below.

Table 1 Age-standardized rates (world) for Asian Indian/Pakistani for the SEER registries of San Francisco, Seattle, Detroit, Atlanta, Connecticut, Los Angeles, and San Jose 1988–2004, SIRs relative to U.S. Whites, ASRs (world) for Bombay and Madras, India 1993–1997

Cancer	Asian Indian/Pakistani age-standardized rate World (n)		SIR (95% CI) Asian Indian/Pakistani relative to U.S. White population		India, Madras 1993–1997 ASR (World Standard)		India, Bombay 1993–1997 ASR (World Standard)	
	Male	Female	Male	Female	Male	Female	Male	Female
All sites	184.88 (2,550)	166.25 (2,466)	0.46 (0.44, 0.48)	0.55 (0.53, 0.58)	106.4	116.6	117.4	124.7
Oral cavity and pharynx	6.73 (108)	3.48 (51)	0.59 (0.46, 0.75)	0.74 (0.51, 1.09)	21.9	10.4	22.8	10.3
Tongue	2.14 (32)	0.87 (14)	0.65 (0.41, 1.04)	0.82 (0.37, 1.76)				
Gum and other mouth	1.81 (33)	1.14 (15)	1.46 (0.83, 2.61)	1.06 (0.48, 2.35)				
Esophagus	2.88 (34)	2.41 (28)	0.50 (0.32, 0.77)	1.77 (0.92, 3.52)				
Stomach	5.76 (77)	3.29 (45)	0.77 (0.56, 1.04)	1.03 (0.67, 1.60)	13.6	6.7	6.5	3.4
Colon and rectum	16.07 (238)	12.65 (173)	0.45 (0.39, 0.53)	0.49 (0.41, 0.59)	5.0	4.1	6.8	5.7
Liver	4.43 (58)	1.96 (23)	0.91 (0.63, 1.32)	1.26 (0.65, 2.48)	2.8	1.1	4.0	2.0
Pancreas	4.77 (60)	2.32 (29)	0.57 (0.41, 0.79)	0.40 (0.25, 0.62)	1.9	0.8	2.5	1.9
Larynx	1.88 (24)	0.67 (7)	0.33 (0.20, 0.53)	0.46 (0.16, 1.19)	4.7	0.5	7.2	1.2
Lung and bronchus	15.50 (202)	6.77 (89)	0.29 (0.25, 0.34)	0.20 (0.15, 0.25)	11.1	2.3	12.3	3.9
Soft tissue	2.39 (40)	1.63 (24)	0.83 (0.53, 1.29)	0.75 (0.42, 1.30)				
Breast	0.52 (9)	59.52 (923)	0.83 (0.30, 2.20)	0.61 (0.56, 0.66)		23.7		30.3
Cervix uteri	NA	3.60 (56)	NA	0.36 (0.26, 0.49)		30.1		17.5
Corpus uteri	NA	10.57 (146)	NA	0.57 (0.46, 0.69)		2.5		2.9
Ovary	NA	9.00 (141)	NA	0.86 (0.68, 1.08)		5.6		8.5
Prostate	63.72 (752)	NA	0.54 (0.49, 0.59)	NA	4.9		7.8	
Urinary bladder	9.19 (116)	3.01 (38)	0.36 (0.29, 0.45)	0.47 (0.31, 0.70)	2.7	1.0	4.8	1.4
Kidney/renal pelvis	5.08 (75)	2.44 (35)	0.44 (0.33, 0.58)	0.45 (0.29, 0.67)	1.2	0.8	2.2	1.2
Brain	4.16 (76)	4.16 (64)	0.62 (0.46, 0.83)	0.84 (0.59, 1.18)	3.0	1.7	3.5	2.4
Thyroid	2.24 (46)	6.20 (127)	0.69 (0.46, 1.01)	0.69 (0.55, 0.87)	1.1	1.6	0.8	2.0
Hodgkin's lymphoma	1.66 (38)	1.63 (29)	0.49 (0.32, 0.73)	0.51 (0.31, 0.81)	1.3	0.5	0.9	0.5
Non-Hodgkin's lymphoma	9.60 (160)	7.84 (123)	0.50 (0.41, 0.61)	0.76 (0.60, 0.97)	3.4	2.2	4.6	3.3
Myeloma	3.96 (48)	2.61 (32)	0.84 (0.56, 1.25)	0.95 (0.57, 1.58)				
Leukemia	9.75 (156)	6.03 (91)	0.84 (0.67, 1.04)	0.83 (0.62, 1.11)	3.9	2.7	4.6	3.6
Lymphocytic leukemia	4.85 (79)	2.61 (38)	0.81 (0.60, 1.10)	0.72 (0.46, 1.12)				
Acute lymphocytic leukemia	2.41 (45)	1.65 (25)	1.13 (0.72, 1.78)	0.91 (0.51, 1.63)				
Acute myeloid leukemia	2.93 (42)	2.39 (36)	0.87 (0.56, 1.34)	1.07 (0.65, 1.76)				
Chronic myeloid leukemia	1.19 (24)	0.61 (11)	0.94 (0.52, 1.71)	0.80 (0.33, 1.90)				

Table 2 Relative risks (RR) and 95% CI from Cox proportional hazards models: SEER 1988–2004

Site		*n*	Cause-specific demographic		Cause-specific demographic + disease		All-cause demographic		All-cause demographic + disease	
			RR (95% CI)	*p*	RR (95% CI)	*p*	RR (95% CI)	*p*	RR (95% CI)	*p*
Lung	Asian Indian/Pakistani Chinese	454	.84 (.74, .95)	.006	.81 (.71, .92)	.001	.89 (.79, .99)	.038	.85 (.76, .95)	.005
		4,783	.91 (.88, .94)	<.0001	.85 (.82, .88)	<.0001	.90 (.88, .93)	<.0001	.85 (.82, .88)	<.0001
Colon	Asian Indian/Pakistani Chinese	379	.63 (.49, .82)	.0004	.68 (.53, .88)	.003	.69 (.57, .84)	.0002	.72 (.59, .88)	.001
		3,850	.90 (.85, .96)	.0007	.92 (.86, .98)	.006	.85 (.81, .89)	<.0001	.86 (.82, .90)	<.0001
Rectal	Asian Indian/Pakistani Chinese	214	.69 (.49, .97)	.032	.85 (.60, 1.20)	.37	.85 (.66, 1.10)	.21	.97 (.74, 1.27)	.97
		1,665	.94 (.86, 1.03)	.17	.98 (.89, 1.08)	.68	.87 (.81, .94)	.0003	.90 (.84, .97)	.006
Stomach	Asian Indian/Pakistani Chinese	194	.82 (.64, 1.04)	.093	.81 (.64, 1.04)	.10	.77 (.64, .93)	.007	.76 (.62, .92)	.006
		1,701	.92 (.86, .98)	.015	.98 (.91, 1.05)	.62	.75 (.71, .80)	<.0001	.81 (.76, .86)	<.0001
Esophagus	Asian Indian/Pakistani Chinese	106	.65 (.49, .86)	.003	.64 (.47, .88)	.005	.65 (.51, .83)	.0007	.68 (.52, .89)	.006
		380	1.00 (.88, 1.14)	.99	.86 (.74, 1.00)	.047	.93 (.82, 1.04)	.19	.80 (.70, .92)	.002
Pancreas	Asian Indian/Pakistani Chinese	132	.79 (.63, 1.00)	.045	.90 (.70, 1.16)	.42	.86 (.70, 1.05)	.14	.97 (.78, 1.22)	.82
		791	.95 (.87, 1.03)	.18	.97 (.89, 1.06)	.51	.96 (.89, 1.03)	.26	.98 (.90, 1.06)	.60
Brain	Asian Indian/Pakistani Chinese	191	.84 (.67, 1.05)	.12	.86 (.68, 1.10)	.22	.91 (.74, 1.11)	.34	.94 (.76, 1.15)	.54
		358	.76 (.66, .89)	.0005	.74 (.63, .87)	.0002	.83 (.73, .95)	.006	.81 (.70, .93)	.003
Liver	Asian Indian/Pakistani Chinese	108	.87 (.64, 1.18)	.36	.89 (.67, 1.17)	.39	.84 (.66, 1.07)	.15	.83 (.64, 1.08)	.17
		1,321	.84 (.78, .92)	<.0001	.83 (.77, .89)	<.0001	.77 (.72, .82)	<.0001	.79 (.73, .85)	<.0001
Female breast	Asian Indian/Pakistani Chinese	1,412	1.16 (.98, 1.37)	.08	.85 (.72, 1.02)	.076	1.20 (1.05, 1.37)	.007	.97 (.85, 1.12)	.97
		5,482	.97 (.89, 1.05)	.40	.93 (.85, 1.01)	.09	.84 (.79, .89)	<.0001	.80 (.75, .85)	<.0001
Ovarian	Asian Indian/Pakistani Chinese	210	1.16 (.91, 1.47)	.22	1.11 (.86, 1.43)	.41	1.24 (1.00, 1.52)	.046	1.19 (.95, 1.48)	.13
		732	.84 (.74, .95)	.006	.94 (.82, 1.08)	.37	.85 (.76, .95)	.005	.95 (.84, 1.07)	.41
Prostate	Asian Indian/Pakistani Chinese	1,207	1.22 (.95, 1.57)	.11	1.00 (.78, 1.28)	.99	.90 (.77, 1.04)	.15	.84 (.72, .98)	.029
		4,557	.80 (.71, .89)	<.0001	.63 (.56, .70)	<.0001	.76 (.72, .81)	<.0001	.70 (.66, .711)	<.0001
NHL	Asian Indian/Pakistani Chinese	416	1.06 (.86, 1.32)	.57	1.11 (.89, 1.37)	.37	.87 (.73, 1.03)	.11	.89 (.75, 1.07)	.21
		1,652	1.17 (1.07, 1.28)	.0006	1.22 (1.12, 1.34)	<.0001	.93 (.86, .99)	.033	.94 (.87, 1.01)	.079

Demographic variables: age, age^2, gender, year of diagnosis, registry

Disease variables: SEER historic stage, tumor grade, histology, tumor size; Except for NHL: tumor grade, histology, extent of disease

Lung and bronchus

Asian Indians/Pakistanis were significantly more likely than Whites to have adenocarcinomas (46.0 vs. 35.4%) which had more favorable prognoses. They were also significantly more likely to have grade I and II, and less likely to have grade III and IV tumors. They were non-significantly more likely to have stage IV disease, and there was no significant difference in tumor size. Asian Indians/Pakistanis had modestly but significantly better survival than Whites in all models. They did slightly better than Chinese in cause-specific models and about the same in all-cause models.

Colon

Compared with Whites Asian Indians/Pakistanis were significantly more likely to have adenomas (46.8 vs. 33.7%). They were non-significantly more likely to have larger tumors, and the grade and stage distributions were very similar. Asian Indians/Pakistanis had significantly better survival than Whites in all models with RRs ranging from .63 to 72 in the various models, with a larger advantage for cause-specific mortality. Adjustment for disease characteristics only slightly attenuated the observed advantage. Chinese also had significantly better survival than Whites but their advantage was less than Asian Indians/Pakistanis, especially for cause-specific survival.

Rectum

Compared with whites Asian Indians/Pakistanis were significantly more likely to have adenomas (50.5 vs. 36.4%), and they had a more favorable stage distribution, 49.1 vs. 43.8% stage 1, and 10.7 vs. 15.1% stage 4, but the difference was not significant ($p = .17$). Size and grade distributions were similar. Asian Indians/Pakistanis had significantly better cause-specific survival but after adjustment for disease characteristics the difference was attenuated to non-significance. Their all-cause mortality was non-significantly better than Whites, but there was almost no difference observed after adjustment for diseases traits. Chinese enjoyed modest advantages in all-cause mortality but their cause-specific mortality was similar to Whites.

Stomach

There were no differences in the distributions of stage, histology, grade, or tumor size between Asian Indians/Pakistanis and Whites. All-cause survival was significantly better for Asian Indians/Pakistanis, while their cause-specific survival was non-significantly better. Adjustment for disease characteristics resulted in little change in the RRs. Compared with Whites, Chinese also had significantly better all-cause survival which was similar to that for Asian Indians/Pakistanis, but had only a slight advantage in cause-specific survival.

Esophagus

Compared with Whites, Asian Indians/Pakistanis were significantly more likely to have squamous cell neoplasms, which had a somewhat less favorable prognosis, and less likely to have adenocarcinomas. Their profile in terms of other tumor characteristics was mixed as they were more likely to have regional stage disease and grade II tumors, but less likely to have stage I or stage IV disease and grade I or grade III/IV tumors. Asian Indians/Pakistanis had significantly and substantially better survival than Whites with RRs below 0.68 in all the models. Chinese had better survival than Whites only after adjusting for disease severity variables.

Pancreas

Asian Indians/Pakistanis were significantly less likely to be diagnosed with distant stage tumors (45.4 vs. 55.7% for Whites). There was little difference between the groups in other disease severity variables. Asian Indians/Pakistanis had marginally significantly better cause-specific survival but this was attenuated to non-significance after adjustment for disease severity. They also had modest and non-significant improved all-cause survival which was also attenuated after adjustment for disease characteristics. There was little difference between Chinese and Whites in any of the models.

Liver

Asian Indians/Pakistanis were slightly and non-significantly more likely to be diagnosed with stage II disease and less likely with stage IV disease. They were also non-significantly more likely to have grade I or III tumor and less likely to have grade II or IV tumor. There was little difference between the groups in tumor size or histology. Asian Indians/Pakistanis showed a modest and non-significant survival advantage relative to Whites. Survival for Chinese was only slightly better than Asian Indians/Pakistanis in all models and was significantly better than White survival (mostly due to larger sample size).

Female breast

Asian Indians/Pakistanis were significantly more likely to be diagnosed with stage II (38.6 vs. 29.9% for whites) and stage IV (6.4 vs. 5.4%) tumors (p for trend $< .0001$). They were also significantly more likely to have grade III and less likely to have grade I tumors and significantly more likely to have larger tumors. There was little difference in histology and more than 90% of both groups had ductal and lobular neoplasms. When adjusting for demographic variable only Asian Indians/Pakistanis had significantly worse all-cause survival and non-significantly ($p = .080$) worse cause-specific survival. After additional adjustment for disease characteristics Asian Indians/Pakistanis had non-significantly ($p = .076$) better survival, and all-cause survival similar to Whites. Chinese women showed a substantial all-cause survival advantage, but had cause-specific survival similar to Whites.

Ovarian

Asian Indians/Pakistanis were slightly and non-significantly less likely than Whites to have stage II or IV tumors. They were significantly more likely to have larger tumors ($p = .008$). Histology and grade distributions were similar for the two groups. Compared with Whites, Asian Indians/Pakistanis had significantly worse all-cause survival after adjustment for demographic factors, and non-significantly worse all-cause survival. Adjusting for disease characteristics attenuated the all-cause results to non-significance. Chinese had significantly better survival in both models adjusting for demographic factors but this was attenuated to non-significance after adjustment for disease-related factors.

Prostate

Asian Indians/Pakistanis were less likely than Whites to have grade 1 tumor (3.4 vs. 7.4%) and more likely to have grade 2 tumor (66.6 vs. 63.5%) ($p < .0001$). They were also more likely to have larger tumors although tumor size was missing for most observations. There was no difference in percentage with stage IV disease. Asian Indians/Pakistanis had a modest and non-significant ($p = .11$) increase in cause-specific mortality relative to Whites which disappeared (RR = 1.00) after adjustment for disease characteristics. They also had non-significantly better all-cause survival which became significant after adjustment for disease variables. Chinese had substantially and significantly better survival in all the models.

Brain

Asian Indians/Pakistanis were less likely than Whites to have glioblastomas (42.4 vs. 49.8%) which have the least favorable prognosis. There was little difference between the groups in grade or size distribution. The survival advantage for Asian Indians/Pakistanis was modest and non-significant and was slightly greater for cause-specific mortality. Chinese showed a greater and significant survival relative to Whites for all the models.

Non-Hodgkin's lymphoma

Asian Indians/Pakistanis were more likely than Whites to have mature t- and nk-cell lymphomas (13.8 vs. 7.2%), which had more favorable prognoses, and less likely to have mature b-cell lymphomas or malignant lymphomas, NOS or diffuse, both of which had less favorable prognoses. They were also more likely to have extranodal NHL, which also had more favorable prognoses. There were no significant differences in grade or extension of disease. Asian Indians/Pakistanis had slightly and non-significantly worse cause-specific survival and slightly and non-significantly better all-cause survival compared with Whites, which showed little changes after adjustment for disease factors. Chinese had modestly and significantly worse cause-specific survival, and slightly better all-cause survival.

Discussion

Our study found that Asian Indians/Pakistanis in the U.S. SEER areas had substantially lower incidence rates for cancers of most sites, with particularly low incidences (SIRs less that 0.50 compared to Whites) for cancers of the lung, larynx, colorectum, urinary bladder, and kidney and renal pelvis in males; lung, colorectum, pancreas, larynx, cervix, bladder, and kidney and renal pelvis in females. Substantial excesses were observed only for gum and other mouth cancer in males, and esophageal cancer in females. In both cases, these excesses were statistically significant if CIs for the SIRs were calculated using standard methods but not significant if the "errors-in-variables" approach was used. In addition, South Asians had generally better survival than Whites for most sites, and the differences were significant for lung, colon, and esophageal cancer in all models, and rectal and pancreatic cancer in some models. In general, survival for Asian Indians/Pakistanis relative to other groups was better for cancers of the digestive system, and worse for gender specific cancers, whereas Chinese generally did better for the gender-specific cancers. Also Chinese tended to do relatively better for all-cause survival than for cause-specific survival, whereas no such pattern was noted for Asian Indians/Pakistanis. Relatively poor breast cancer survival for Asian Indian/Pakistani females appears to result from their tendency to present with more advanced disease. A study of breast cancer screening practices in Asian-Americans found that only 5% of Asian Indian women practiced monthly breast self-examination, as compared with 23 and 51% of Chinese and Filipino women respectively [13]. Surveys from the United States have also reported Asian Indian women were less likely to have had a mammogram within the preceding two years [15] and more likely to report never having had another mammogram than Whites or other Asian-American groups [16].

Our incidence results are generally consistent with a study using data from the California cancer registry [1], and indeed there is overlap between the studies because both would have included cases during San Francisco during 1988 to 2000, and San Jose and Los Angeles during 1991 to 2000. Our incidence rates are generally lower than theirs, particularly for women, but our male oral cavity and lung cancer rates were higher than theirs. Also in contrast with their study, we did not find a substantially higher liver cancer rate for Asian Indians/Pakistanis relative to the White population from the same geographical areas. The differences between the studies could result from geographical differences between Asian Indian/Pakistani populations in different parts of the United States and from temporal differences as our study included more recent data during 2001 to 2004. A study from the United Kingdom [2] also found significantly lower rates for south Asians overall, and for several important sites including the lung, colon, rectum, and female breast. Their study also found, as did ours that while south Asian males had lower esophageal cancer rates, south Asian females had higher rates than non-south Asians. The overall magnitude of the reduced risk for south Asians relative to non-South Asians from their study was generally lower than for our study. Differences are not surprising as the South Asian population of the United Kingdom differs from that of the United States in terms of national origin, with a larger percentage coming from Pakistan and Bangladesh [17], and SES, since the Pakistani and Bangladeshi segments of the South Asian population in the United Kingdom tend to be relatively poor [17].

One contributing factor to lower cancer incidence in Asian Indians/Pakistanis is undoubtedly lower smoking prevalence, especially for females. A survey conducted during 1999 to 2001 in the United States found that 20 and 3% of Asian Indian adult males and females smoked compared to 29 and 26% for White males and females respectively [18]. The differences in smoking prevalence are considerably stronger for females than for males and this is reflected in the lower SIR for lung cancer for Asian Indian females (0.20) than for males (0.29). However, for other smoking-related cancers such as bladder, stomach, and particularly esophageal cancers the SIRs for Asian Indian females are higher than those for males. This may be due to the fact that smoking is a considerably more dominant risk factor for lung cancer than for these other cancers [19] and thus other lifestyle or genetic factors may have a stronger influence on gender and ethnic differences in the incidence of these other cancers. The lower smoking prevalence among Asian Indians may also be a contributing factor to their generally better cancer survival as smoking may worsen the prognosis for certain cancers [20, 21] as well as increasing incidence. Dietary factors are also likely important. Diets on the Indian subcontinent generally consist of a greater quantity and variety of vegetables and lower meat consumption [22, 23]. Curcumin, an ingredient used in curry powders on the Indian subcontinent, has been shown to protect against carcinogenesis and metastasis in a number of animal tumor models [24] Whether or not the amount of curcumin found in turmeric or curry powders is enough to have an impact on cancer risk is not known [25]. Another likely contributing factor to their lower cancer incidence is the lower prevalence of obesity, a risk factor for several cancers, in the U.S. Asian Indian population. A study using data from the U.S. National Health Interview Survey during 1997 to 2000 found that the prevalence of obesity among Asian Indians was 6.2 vs. 19.2% for non-Hispanic whites [26]. Given their higher average SES they may also have less exposure to occupational risk factors for cancer.

Asians or Asian/Pacific Islanders are often treated as a single group in epidemiological research in the United States despite the fact that they are a diverse group in terms of culture, length of stay in the United States, and socioeconomic status. Inclusion of Asian Indians and Pakistanis in this group seems even less appropriate as they are genetically more similar to Caucasians than to East Asians [27]. According to U.S. Census data Asian Indians are the wealthiest major ethnic group in the country, with a median household income in 1999 of US$63,669 per year, as against $44,687 for Whites, $51,444 for Chinese, $60,370 for Filipinos, and $47,241 for Pakistanis [28]. The extent, to which their apparent survival advantage for many cancers can be attributed to better treatment, cannot be determined from this study, but the advantage does not generally appear to be due to earlier diagnosis or more favorable tumor characteristics.

Most Asian Indians and Pakistanis in the United States are recent immigrants. According to the 2000 U.S. Census data about only about 9% of Asian Indians and 6% of Pakistanis over the age of 18 were born in the U.S. [29]. These data also show that the majority of adults in both groups immigrated after 1985 [29], thus that the extent of acculturation to the United States is likely to be quite low. The patterns of cancer incidence for these groups may begin to more closely resemble those of the general U.S. population as they become more acculturated over time.

Our study does have some limitations. The SEER data lumps Asian Indians and Pakistanis into one category so there is no way to draw separate inferences for these groups which are culturally very different. According to 2000 census data, Asian Indians made up about 94% of the Asian

Indian/Pakistani population in the SEER areas used for incidence and for survival so our inferences primarily apply to this group [30]; but there is no way of knowing how different the groups are from each other in terms of cancer incidence and survival. Other "South Asian" groups including Bangladeshis and Sri Lankans are classified as "other Asian" by SEER. The SEER registries used for the incidence analysis cover about 19% of the Asian Indian population and 12% of the Pakistani population of the United States, while those used for the survival analyses covered 37 and 28% of these populations [30]. As always, there is also the possibility of racial misclassification because race was abstracted from medical records, and some Asian Indians and Pakistanis strongly resemble Europeans physically. We feel that any bias resulting from this should be non-differential. Despite these limitations, we feel that the strengths of this study, including its population-based nature, use of a geographically diverse cohort, and the large sample size make the study findings accurate and relevant.

In conclusion, our review of the U.S. SEER data has found that Asian Indians/Pakistanis living in the United States have a low incidence of most major cancers compared to U.S. Whites but a higher incidence than what is found in India. We have also found that they generally have better survival rates following cancer diagnosis, particularly for cancers of the digestive system, and that this improved survival does not appear to result from more favorable disease characteristics at diagnosis. Further research is necessary to identify dietary and lifestyle factors, other than smoking, responsible for the low cancer incidence in this group, and to determine whether their better-than-average cancer survival is explained by their high relative SES and possibly better treatment options.

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